ECO 4000 — Midterm Mock Exam
Spring 2026

Write your name and ID eligibly please: _____

You have 20 questions, please read each question carefully twice. All questions are weighted equally. Do not make stupid mistakes. You have 75 minutes. "RV" = random variable. $N(a, b)$ = normal distribution with mean a and variance b. Z-table and t-table are on the last page. Clearly mark your answers on Bubble sheet. You ARE allowed to use cheat sheet. You ARE NOT allowed to use any other materials or devices. You cannot use or copy from other student cheat sheets. Cheating or any other dishonest behavior will be punished by ZERO points for the exam.

GOOD LUCK!

1. Which of the following is an example of cross-sectional data?
   (a) Monthly unemployment rates in the United States from 2000 to 2023.
   (b) Quarterly revenue of a single firm from 2010 to 2022.
   (c) The GDP of every country in the world measured in the year 2022.
   (d) Annual earnings of a panel of workers tracked over 10 years.

2. A school district wants to test whether smaller class sizes raise test scores. It randomly assigns students to classes of 15 or 25 students and compares scores at the end of the year. What makes this a randomized controlled experiment?
   (a) The outcome variable (test scores) is continuous.
   (b) Random assignment ensures that differences in scores are attributable to class size rather than pre-existing differences between students.
   (c) The sample is drawn from the full population of all schools.
   (d) The study uses two groups, one treated and one untreated.

3. Let X be a RV with mean 3 and variance 4. What is $E(X^2)$?
   (a) 4
   (b) 9
   (c) 13
   (d) 7

4. Let M be a RV with mean -2 and variance 9. Let $G = -3M + 2$. What is $E(G)$?
   (a) -4
   (b) 4
   (c) 8

(d) 2

5. The correlation between X and Y is $Corr(X, Y) = 0.5$, $SD(X) = 4$, and $SD(Y) = 3$. What is $Cov(X, Y)$?

   (a) 2.0
   (b) 6.0
   (c) 0.5
   (d) 12.0

6. Which of the following statements about the correlation coefficient is NOT correct?

   (a) $Corr(X, Y)$ always lies between -1 and 1.
   (b) $Corr(X, Y) = Corr(Y, X)$
   (c) $Corr(X, Y) > 0$ implies that X and Y are independent.
   (d) $|Corr(X, Y)| = 1$ implies a perfect linear relationship between X and Y.

7. Let $X \sim N(10, 25)$. What is $P(X < 15)$?

   (a) 0.5000
   (b) 0.6915
   (c) 0.8413
   (d) 0.9772

8. Using the same distribution $X \sim N(10, 25)$, a student claims: "$P(10 < X < 15) = 0.3413$, because by symmetry it is exactly half of $P(5 < X < 15) = 0.6826$." Is the student correct?

   (a) No; the correct probability is 0.1587.
   (b) No; the correct probability is 0.6826.
   (c) No; the normal distribution cannot be split this way.
   (d) Yes; by the symmetry of the normal distribution around its mean ($\mu = 10$), $P(10 < X < 15) = 0.3413$.

Questions 9–13 refer to the following scenario. A researcher surveys $n = 36$ workers and records weekly wages. The sample mean is $\overline{Y} = \$78$ and the sample standard deviation is $s = \$12$.

9. The researcher reports $SE(\overline{Y}) = 2$. Which of the following best describes what this standard error measures?

   (a) The standard deviation of individual wage observations in the sample.
   (b) The difference between the sample mean and the true population mean.
   (c) The approximate standard deviation of the sample mean $\overline{Y}$ across many repeated samples of size 36.

(d) The width of the 95% confidence interval.

10. What is the 95% confidence interval for the population mean weekly wage?

    (a) [\$74.08, \$81.92]
    (b) [\$70, \$86]
    (c) [\$76, \$80]
    (d) [\$74.40, \$81.60]

11. You test $H_0 : \mu = 80$ vs. $H_1 : \mu \neq 80$. What is the t-statistic?

    (a) $t = -2$
    (b) $t = 1$
    (c) $t = -1$
    (d) $t = 2$

12. For which of the following sample sizes is it most important to use the t-table instead of the z-table for hypothesis testing?

    (a) $n = 500$
    (b) $n = 15$
    (c) $n = 200$
    (d) $n = 1,000$

13. What is the approximate two-sided p-value for the test in question 11? (Hint use the Z-table)

    (a) 0.0794
    (b) 0.1587
    (c) 0.8413
    (d) 0.3174

14. Based on the p-value from question 13, what is the correct conclusion at the 5% significance level?

    (a) Reject H_0; the p-value is less than 0.05.
    (b) Fail to reject H_0; the observed sample mean is consistent with $\mu = 80$.
    (c) Reject H_0; the sample mean differs from 80.
    (d) Cannot conclude anything without knowing the population variance.

Questions 15–20 refer to the following OLS regression. Weekly Wage (in \$) is regressed on Years of Education (Edu):

$$\widehat{\text{Wage}} = \underset{(3)}{8} + \underset{(0.5)}{2.5} \times \text{Edu}$$

$$R^2 = 0.50, \quad SER = 4$$

(Standard errors in parentheses)

15. A classmate argues: "$\hat{\beta}_1 = 2.5$ proves that completing one extra year of education causes weekly wages to increase by \$2.50." What is the correct response?

    (a) The classmate is right; OLS is designed to estimate causal effects.
    (b) The classmate is right; the coefficient is statistically significant.
    (c) The classmate is wrong; $\hat{\beta}_1 = 2.5$ describes a correlation, not necessarily causation. Omitted variables (e.g., ability) could be driving the relationship.
    (d) The classmate is wrong; the slope should be divided by the standard error first.

16. What is the difference in predicted weekly wage between a worker with 16 years of education and one with 12 years?

    (a) \$2.50
    (b) \$7.50
    (c) \$10.00
    (d) \$40.00

17. What is the t-statistic for the two-sided test of $H_0 : \beta_1 = 0$?

    (a) $t = 0.2$
    (b) $t = 5.0$
    (c) $t = 2.5$
    (d) $t = 1.6$

18. The p-value for the test $H_0 : \beta_1 = 0$ is approximately 0.0000. Which statement is correct?

    (a) Education has no effect on wages.
    (b) We reject $H_0 : \beta_1 = 0$ at all conventional significance levels.
    (c) The regression explains all variation in wages.
    (d) The slope is economically large.

19. In this regression, $SSR = 800$ and $TSS = 1,600$. What is R^2?

    (a) 0.40
    (b) 0.44
    (c) 0.50
    (d) 0.60

20. Using the SER and sample size from this regression ($SER = 4$, $n = 52$), what is the Sum of Squared Residuals (SSR)?

    (a) $SSR = 800$
    (b) $SSR = 200$
    (c) $SSR = 16$
    (d) $SSR = 8,000$

$$\Phi(x) \approx \Phi(x_1) + \frac{x - x_1}{x_2 - x_1}\left(\Phi(x_2) - \Phi(x_1)\right)$$

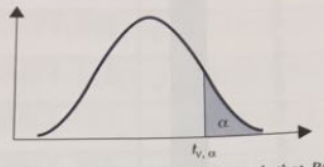
x	$\Phi(x)$	x	$\Phi(x)$	x	$\Phi(x)$	x	$\Phi(x)$	x	$\Phi(x)$	x	$\Phi(x)$
0.00	0.5000	**0.50**	0.6915	**1.00**	0.8413	**1.50**	0.9332	**2.00**	0.9772	**2.50**	0.9938
0.05	0.5199	**0.55**	0.7088	**1.05**	0.8531	**1.55**	0.9394	**2.05**	0.9798	**2.55**	0.9946
0.10	0.5398	**0.60**	0.7257	**1.10**	0.8643	**1.60**	0.9452	**2.10**	0.9821	**2.60**	0.9953
0.15	0.5596	**0.65**	0.7422	**1.15**	0.8749	**1.65**	0.9505	**2.15**	0.9842	**2.65**	0.9960
0.20	0.5793	**0.70**	0.7580	**1.20**	0.8849	**1.70**	0.9554	**2.20**	0.9861	**2.70**	0.9965
0.25	0.5987	**0.75**	0.7734	**1.25**	0.8944	**1.75**	0.9599	**2.25**	0.9878	**2.75**	0.9970
0.30	0.6179	**0.80**	0.7881	**1.30**	0.9032	**1.80**	0.9641	**2.30**	0.9893	**2.80**	0.9974
0.35	0.6368	**0.85**	0.8023	**1.35**	0.9115	**1.85**	0.9678	**2.35**	0.9906	**2.85**	0.9978
0.40	0.6554	**0.90**	0.8159	**1.40**	0.9192	**1.90**	0.9713	**2.40**	0.9918	**2.90**	0.9981
0.45	0.6736	**0.95**	0.8289	**1.45**	0.9265	**1.95**	0.9744	**2.45**	0.9929	**2.95**	0.9984
0.50	0.6915	**1.00**	0.8413	**1.50**	0.9332	**2.00**	0.9772	**2.50**	0.9938	**3.00**	0.9987

The inverse function $\Phi^{-1}(p)$ is tabulated below for various values of p.

p	**0.900**	**0.950**	**0.975**	**0.990**	**0.995**	**0.999**	**0.9995**
$\Phi^{-1}(p)$	1.2816	1.6449	1.9600	2.3263	2.5758	3.0902	3.2905

Figure 1: Standard Normal (Z) Distribution Table

TABLE 6 Cutoff points for the Student's t distribution



For selected probabilities, α, the table shows the values $t_{\nu,\alpha}$ such that $P(t_\nu > t_{\nu,\alpha}) = \alpha$, where t_ν is a Student's t random variable with ν degrees of freedom. For example, the probability is .10 that a Student's t random variable with 10 degrees of freedom exceeds 1.372.

ν	α				
	.100	.050	.025	.010	.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
40	1.303	1.684	2.021	2.423	2.704
60	1.296	1.671	2.000	2.390	2.660
∞	1.282	1.645	1.960	2.326	2.576

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Figure 2: Student's t-Distribution Table